

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY, LONERE – RAIGAD - 402 103
Semester Winter Examination – December, 2019**

Branch: B. Tech in Information Technology
Subject: Internetworking Protocols (BTITC404)
Date: -02/12/2019

Sem.: - IV
Marks: 60
Time: - 3 Hrs.

57

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc. wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

- Q.1. A) Explain layers in the TCP/IP protocol suite. 06
B) What is Gigabit Ethernet? Show its implementations. 06
- Q.2. A) Distinguish between direct and limited broadcast special addresses with example networks. 06
B) Explain the fields in IEEE 802.11 MAC frame. 06
- Q.3. A) Explain the fields which are related to fragmentation and reassembly of an IP datagram. 06
B) Illustrate with figure, four different cases which use the services of ARP (Address Resolution Protocol). 06
- Q.4. A) How the *traceroute* program can be used to trace route of a packet from the source to the destination? 06
B) Write and explain distance vector routing algorithm. 06
- Q.5. A) Explain checksum calculation in UDP (User Datagram Protocol). 06
B) Draw simplified UDP package diagram, showing its components and their interactions. Write Input Module. 06
- Q.6. A) Explain three way handshaking in TCP (Transmission Control Protocol) connection establishment. 06
B) What are the most common fields in TCP's Transmission Control Blocks (TCBs)? 06

Paper End